

COMPUTER SCIENCE 2025

(Science General Group)

Option-I Programming using 'C'**TIME: 3 Hours****(75 Marks)****SECTION 'A' (Multiple Choice Questions)(15)****NOTE:** i) This section consists of 15 part questions and all are to be answered. Each part questions carries 1 mark.ii) Do not copy the part questions in your answerbook. Write only the answer in full against the proper number of the question and its part.

1 Choose the correct answer for each from the given options:

i) The rules/formats for using a programming language are called:

*semantic *rule *syntax✓ *module

ii) Variable and constant name cannot contains a(n):

*Letter *Underscore *Period✓ *Number

iii) This is a binary operator:

*? *++ *-- *+✓

iv) Every C-statement must be ends with:

*Period *Comma *Colon *Semi colon✓

v) The number of bytes used by "float" data type in C is:

*2 *4✓ *6 *8

vi) This functions is used to get input from the user:

*paintf() *scanf()✓ *clrscr() *puts()

vii) $a += b$ is equivalent to:* $b += a$ * $a = +b$ * $a = a + b$ ✓ * $b = b + a$

viii) These are logical operators except:

*&& *|| *! *>=✓

ix) This function returns the length of the strings:

*strep() *streat() *stricon() *strlen()✓

x) The escape sequence for new line is:

*\ a *\\ n✓ *\\ r *\\ f

xi) This one is not a loop structure:

*Switch ✓ *for *while *do-while

xii) The symbol " = " represents:

*comparison operator *Equal to operator
*Assignment operator *Logical operator

xiii) The array int marks [] = {500, 700, 600, 800, 900}, the marks [2]

is:
*500 *600✓ *700 *800

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- x) A pictorial representation of program logic is called:
 * Flow chart ✓ * Procedure
 * Source code * Algorithm
- x) Loop within the loop is called:
 * Nested loop ✓ * Infinite loop
 * Inner loop * Outer loop
- xi) The C pre-processors directives specified with this symbol:
 * # ✓ * % * \$ * &
- xii) Every C program must contains:
 * Atleast one function ✓ * No function
 * No value from command line * multiple functions
- xiii) In database, columns are called:
 * Toples * Field ✓ * Keys * Files
- xiv) scanf() is predefined function in this header file:
 * stdlib.h * stype.h * stdio, h ✓ * stdrag.h
- xv) The format specifier % f is used for:
 * int * char * float ✓ * double

SECTION "B" (Short Answer Question) Marks: 30

NOTE: Attempt any TEN part questions in all. All part questions carry equal marks. Answer should not exceed 6 to 7 lines

2. i) What is meaning of two void in any function declaration? i.e void ABC(void);

ii) Differentiate between any one.

a) Source Code and Object Code

b) Variable and Local Variable

iii) What are pre-processor directives in C language? Name any two:

iv) Write a program to display the following output.

```
6    12    18
7    14    21
8    16    24
```

v) What will be the output of the C code.

```
main()
{
    int a = 11;
    while(a <= 20) {
        printf("%d", a);
        a += 2;
    }
}
```

vi) Convert the following, mathematical expression into C expression:

$$a. Y = (x + y) - (a^2 + b^3) \quad b. c = \sqrt{x + y + z} \quad c. P = \frac{(a^2 + b^2)}{(a - b)^2}$$

vii) Define any two input and output functions in C.

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- viii) Differentiate between while loop and do while loop.
- ix) What is the purpose of the main() function in any C program?
- x) What is the function of "break" statement in C?
- xi) What is a DBMS?
- xii) Define primary key and foreign key
- xiii) Write the full form of any three of the following acronyms:
 * DML * DDL * SQL * IDE * RDBMS
- xiv) Write a program in C to check whether the entered number is even or odd:
- xv) What is the role of DBA in DBMS?

Section C (Detailed - Answer Question) (30)

NOTE: Attempt any THREE question from this section.

All questions carry equal marks.

- What is user-defined function? How is it differ from library function? Write any two with examples.
- What is relationship in database? Define three types of relationships in database with examples.
- Explain various types of database models.
- What is control structure? Explain selection control structure.
- Explain all the data types in C language.

COMPUTER 2023

Option-I Programming using 'C'

TIME: 3 Hours

(75 Marks)

SECTION 'A' (Multiple Choice Questions)(15)

1 Choose the correct answer for each from the given options:

(i) A language that differentiates between upper-case and lower-case is called:

- * Machine Language * Visual Language
 * Case Sensitive ✓ * High Level Language

(ii) This symbol is used for C preprocessor directive:

- * # ✓ * & * @ * %

(iii) These cannot be declared as identifiers:

- * Global Variable * Local Variable
 * Reserve Words * Formal Parameter

(iv) Only this special character is allowed in identifier:

- * _ (underscore) ✓ * & * @ * &&

(v) Escape sequence character is followed by:

- * \ ✓ * / * % * {

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- (vi) This function only inputs string type data:
 * scanf() * gets() * getch() * getche()
- (vii) This is an address operator:
 * && * * * & * ~
- (viii) This statement suspends the control structure:
 * continue * break * getch() * default
- (ix) This statement terminates the loop:
 * break * continue * getch() * return()
- (x) This is correct input statement to take 'gender as char':
 * getch(&gender) * gender=getche()
 * getche()-gender * gender()-getche()
- (xi) Variables defined within function definition are called:
 * Actual Parameters * Formal Parameters
 * Global Variables * Local Parameters
- (xii) It is an organized collection of related data, typically stored on disk and accessible by possibly many concurrent users:
 * Source Code * Text File
 * Database * Object File
- (xiii) This database model arrange data in a parent child relationship:
 * Relation Model * Network Model
 * Hierarchical Model * Flat Model
- (xiv) This database model may have more than one parents:
 * Relation Model * Network Model
 * Hierarchical Model * Flat Model
- (xv) In database columns are called:
 * Tuples * Attributes * Keys * File

SECTION "B" (Short Answer Question) Marks: 30

NOTE: Attempt any TEN part questions in all. All part questions carry equal marks. Answer should not exceed 6 to 7 lines

- (i) Why C language is called case-sensitive language? Give an example
- (ii) What is C-Preprocessor Directive? Define #define OR #include
- (iii) Define the following functions with an example.
 * abs() * pow() * sqrt()
- (iv) Convert the following into C-expressions:

a) $A = \pi r^2$, b) $y = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$ c) $a = \frac{4}{3} \pi r^3$

- (v) Write a program that takes the value from user and determine its factorial.

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- (vi) What is an Address Operator? Give an example.
- (vii) Write a program that takes two numbers and prints the greater number.
- (viii) Write a program that prints the following pattern using nested loop.
 + + + + +
 + + + +
 + + +
 + +
 +
- (ix) Write output of the given code.
 void main (void)
 {
 int i,j;
 for(i = 1; i <= 5, i++)
 {
 for (j = 1; j <= i; j++)
 printf("%d"j*2);
 printf("\n");
 }
 }
- (x) Define two ways in which arguments can be passed to a function.
- (xi) Differentiate between internal/local and external/global variables.
- (xii) Define any three of the following:
 * fopen() * fclose() * fprintf()
- (xiii) What is a Database? Write its advantages.
- (xiv) What are main objects in MS ACCESS?
- (xv) Define Database Relationships. Write its uses.

Section C (Detailed -Answer Question) (30)

NOTE: Attempt any TWO question from this section.

All questions carry equal marks.

- What is an Operator? Explain basic types of operators in C.
- Explain unformatted input and output (I/O) functions in C with syntax and examples.
- Explain switch statement with syntax and an example.
- What is a Loop? Explain while and do loop with examples.
- Explain the database models with diagram.

COMPUTER 2022

OPTION - I - PROGRAMMING USING "C"

TIME: 3 Hours

(75 Marks)

SECTION 'A' (Multiple Choice Questions)(29)

1 Choose the correct answer for each from the given options:

- (i) In $x = 3.5 / a + 2 * n$; this operation will be performed first:
 * $3.5 / a$ ✓ * $a + 2$ * $2 * n$ * depends upon compiler
- (ii) Every function's body begins and ends with:
 * { } ✓ * () * { | } * < >
- (iii) This operator takes only integer operands:
 * + * * % ✓ * /
- (iv) This is not a logical operator:
 * & ✓ * && * || * !
- (v) If $int a = 10 + 4.867$ then the value of 'a' is:
 * $a = 10$ * $a = 14.867$ * $a = 14$ ✓ * compiler error
- (vi) This operator takes only one operand:
 * ! * || * && * >
- (vii) This inversion specifier is used to display a floating point value in exponential notation:
 * [e or E] ✓ * [d or D] * [g or G] * [f or F]
- (viii) This statement can only test for equality:
 * break * switch ✓ * return * if
- (ix) This data type is used when the function does not return a value:
 * char * double * float * void ✓
- (x) The declaration of a function is called its:
 * Prototype ✓ * Function * Function Call * Function Definition
- (xi) We use this operator to tell the compiler to store data at an address:
 * address operator (&) ✓ * dereference operator (#) * indirection operator (*) * pointer operator (^)
- (xii) A function can return only:
 * one values ✓ * two values * three values * six values
- (xiii) This is a default data type in MS-Access:
 * auto number * text ✓ * number * data
- (xiv) This MS Access object is considered primary:
 * table ✓ * query * form * report
- (xv) Key field must contain:
 * default values * null values * unique values ✓ * duplicate values
- (xvi) 'One-to-many' refers to:
 * constraint * join * union * cardinality ✓
- (xvii) This is the output of 4% 5;

- * 0 ✓ * 1 * 4 * 5
- (xviii) If n has the value 3, then the output of the statement, `printf("%d%d", n++, ++n);` is:
 * 3 5 * 4 5 ✓ * 4 4 * 2 5
- (xix) To retrieve information from a database, users can use this, an easy to use computer language:
 * data dictionary * report generator * query language ✓ * low level language
- (xx) The rules/formats for using programming language are called:
 * Sematic * Rule * Syntax ✓ * Module
- (xxi) This statement is used to exit the selection structure:
 * continue * default * break ✓ * exit
- (xxii) If a file is opened in "w" mode, then:
 * Reading is possible * Appending is possible * Writing is possible ✓ * there is no any "w" mode
- (xxiii) Data access must be sequential in:
 * Network Database Model * Hierarchical Database Model * Relational Model ✓ * Flat Model
- (xxiv) "==" is an operator of the type:
 * Assignment * Arithmetic ✓ * Relational * Logical
- (xxv) This is used to share the data from two tables:
 * Table ✓ * Form * Relation * Report
- (xxvi) This data type is used to store long text entries in MS-Access:
 * Memo ✓ * Text * Number * OLE object
- (xxvii) In this database mode, data is organized into related groups much like a family tree:
 * Hierarchical ✓ * Network * Relational * Flat
- (xxviii) RDBMS consists of:
 * Collection of Records * Collection of Keys * Collection of Tables ✓ * Collection of fields
- (xxix) In MS-Access, this object is used to represent data in a printed format:
 * Query * Form * Report ✓ * Macro

SECTION "B" (Short Answer Question) Marks: 30

NOTE: Attempt any TEN part questions in all. All part questions carry equal marks. Answer should not exceed 6 to 7 lines

- 2.i) Why is C language called 'case sensitive language'?
- ii) What will be the output of the following program segment?
- ```

For(i = 1; i < 5; i++)
if (i == 3)
continue;
else
print("%d", i);

```
- iii) What is a function? What are the advantages of using a function?

# COMPUTER 2022

## OPTION - I - PROGRAMMING USING "C"

(75 Marks)

TIME: 3 Hours

### SECTION 'A' (Multiple Choice Questions)(29)

1 Choose the correct answer for each from the given options:

- (i) In  $x = 3.5 / a + 2 * n$ ; this operation will be performed first:  
 \*  $3.5 / a$  ✓ \*  $a + 2$  \*  $2 * n$  \* depends upon compiler
- (ii) Every function's body begins and ends with:  
 \* { } ✓ \* ( ) \* | | \* < >
- (iii) This operator takes only integer operands:  
 \* % ✓ \* / \* \* \* \*
- (iv) This is not a logical operator:  
 \* & ✓ \* && \* || \* ! \* \*
- (v) If  $\text{int } a = 10 + 4.867$  then the value of 'a' is:  
 \*  $a = 10$  \*  $a = 14.867$  \*  $a = 14$  ✓ \* compiler error
- (vi) This operator takes only one operand:  
 \* ! \* || \* && \* > \* \*
- (vii) This inversion specifier is used to display a floating point value in exponential notation:  
 \* [e or E] ✓ \* [d or D] \* [g or G] \* [f or F]
- (viii) This statement can only test for equality:  
 \* break \* switch ✓ \* return \* if
- (ix) This data type is used when the function does not return a value:  
 \* char \* double \* float \* void ✓
- (x) The declaration of a function is called its:  
 \* Prototype ✓ \* Function \* Function Call \* Function Definition
- (xi) We use this operator to tell the compiler to store data at an address:  
 \* address operator(&) ✓ \* dereference operator (#) \* indirection operator(\*) \* pointer operator (^)
- (xii) A function can return only:  
 \* one values ✓ \* two values \* three values \* six values
- (xiii) This is a default data type in MS-Access:  
 \* auto number \* text ✓ \* number \* data
- (xiv) This MS Access object is considered primary:  
 \* table ✓ \* query \* form \* report
- (xv) Key field must contain:  
 \* default values \* null values \* unique values ✓ \* duplicate values
- (xvi) 'One-to-many' refers to:  
 \* constraint \* join \* union \* cardinality ✓
- (xvii) This is the output of 4% 5:

- \* 0 ✓ \* 1 \* 4 \* 5
- (xviii) If n has the value 3, then the output of the statement, `printf("%d%d", n++, ++n);` is:  
 \* 3 5 \* 4 5 ✓ \* 4 4 \* 2 5
- (xix) To retrieve information from a database, users can use this, an easy to use computer language:  
 \* data dictionary \* report generator \* query language ✓ \* low level language
- (xx) The rules/formats for using programming language are called:  
 \* Sematic \* Rule \* Syntax ✓ \* Module
- (xxi) This statement is used to exit the selection structure:  
 \* continue \* default \* break ✓ \* exit
- (xxii) If a file is opened in "w" mode, then:  
 \* Reading is possible \* Appending is possible \* Writing is possible ✓ \* there is no any "w" mode
- (xxiii) Data access must be sequential in:  
 \* Network Database Model \* Hierarchical Database Model \* Relational Model ✓ \* Flat Model
- (xxiv) "++" is an operator of the type:  
 \* Assignment \* Arithmetic ✓ \* Relational \* Logical
- (xxv) This is used to share the data from two tables:  
 \* Table ✓ \* Form \* Relation \* Report
- (xxvi) This data type is used to store long text entries in MS-Access:  
 \* Memo ✓ \* Text \* Number \* OLE object
- (xxvii) In this database mode, data is organized into related groups much like a family tree:  
 \* Hierarchical ✓ \* Network \* Relational \* Flat
- (xxviii) RDBMS consists of:  
 \* Collection of Records \* Collection of Keys \* Collection of Tables ✓ \* Collection of fields
- (xxix) In MS-Access, this object is used to represent data in a printed format:  
 \* Query \* Form \* Report ✓ \* Macro

### SECTION "B" (Short Answer Question) Marks: 30

**NOTE:** Attempt any TEN part questions in all. All part questions carry equal marks. Answer should not exceed 6 to 7 lines

2.i) Why is C language called 'case sensitive language'?

ii) What will be the output of the following program segment?

```

For(i = 1; i < 5; i++)
 if (i == 3)
 continue;
 else
 printf("%d", i);

```

iii) What is a function? What are the advantages of using a function?

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iv) Define any one of the following:

a) DBMS \* Data dictionary

v) Write a program that generates the following output: (use any loop)

```
2 4
3 6
4 8
5 10
```

vi) Write a program that prints the factorial of given number.

vii) What is reserved word or keywords in C language? Define only three.

viii) What are escape sequences? Write the purpose of any three.

ix) Define a) Table b) Query

x) What is data redundancy?

xi) Define increment operator and decrement operator. Give example of each operator.

xii) What is the difference between a primary key and a secondary key?

xiii) Write equivalent C statement:

a)  $x = \frac{b^2 - c^2}{2bc}$  b)  $(x + y)^a$  \*  $b = \sqrt{x + y} + z^2$

ix) Define scanf() with its syntax

x) Differentiate between Actual Parameter and Formal Parameters.

### Section C (Detailed - Answer Question) (16)

**NOTE:** Attempt any TWO question from this section.

All questions carry equal marks.

3. Explain arithmetic, relational and logical operators used in C.

4. Define switch statement. Also write an example program of switch statement.

5. Explain different types of database models.

## COMPUTER 2021

(AS PER CONDENSED SYLLABUS)

### OPTION - I - PROGRAMMING USING "C"

TIME: 2 Hours

(100 Marks)

**NOTE:** i) This section consists of 36 part questions and all are to be answered. Each question carries ONE marks.

ii) Do not copy the part questions in your answer book. Write only the answer in full against the proper number of the question

### SECTION 'A' (Multiple Choice Questions) (36)

1. Select the correct answer for each from the given options:

(i) This data type in C occupies 4 times as many bytes of memory as type char:  
\* int \* float \* double \* long float

(ii) A block of instructions that performs a specific task, is a/an:  
\* function \* loop \* string \* array

## PAST PAPERS XII (S) 403 COMPUTER (C)

(iii) Every statement in C must end with:

\* ; \* ) \* , \* .

(iv) A variable that can hold the value 234 having a data type:

\* char \* float \* int \* double

(v) A relational operator is used to:

\* change variable to logical values \* Compare values

\* distinguish different types of variables \* combine values

(vi) This statement causes an exit only from the inner most switch:

\* continue \* goto \* break \* default

(vii) Multiple increment expression in a for loop expression is separated by:

\* < \* ; \* , \* /

(viii) A string is terminated by a null character which is written as:

\* \0 \* // \* \n \* \t

(ix) Type char can accommodate numbers from:

\* -32768 to +32767 \* 3.4e-38 to 3.4e+38

\* -128 to 127 \* 0 to 65565

(x) The function designed specially to read string from the keyboard:

\* getch() \* puts() \* gets() \* getchar()

(xi) An external variable is defined:

\* in the first function that uses it

\* in any function that uses it

\* in main () only

\* outside of any function

(xii) This function is used to clear the screen:

\* clear \* clrscr \* Clrscr \* CLS()

(xiii) This format specifier is used for exponential notation:

\* %e \* %x \* %xd \* %d

(xiv) In switch statement if no match is found with any case statement, only the statements following are executed in:

\* else \* continue \* default \* break

(xv) A loop performed within another loop is said to be:

\* nested loop \* conditional loop

\* recursive loop \* continuous loop

(xvi) Three types of control structures are:

\* initialization, condition, increment

\* sequence, selection, iteration

\* sequential, selection, conditional

\* initialization, condition, decrement

(xvii) String is:

\* a collection of characters \* a list of characters

\* an array of characters \* a single character

(xviii) This will be output of 22% 3:

\* 3 \* 1 \* 22 \* 0

(xix) The function scanf() reads:

\* any possible number \* a single character

\* any possible variable type \* a character and a string

## PAST PAPERS XII (S)

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## COMPUTER (C)

(xx) This loop will be executed at least once whether its condition is false:

- \* for
- \* while
- \* do-while ✓
- \* continuous loop

(xxi) A variable which can be accessed by all program modules is called:

- \* global variable ✓
- \* local variable
- \* argument
- \* private variable

(xxii) Case statement is used with:

- \* printf()
- \* scanf()
- \* switch () ✓
- \* getch()

(xxiii) This statement is incorrect in C:

- \* int a; float b;
- \* double c;
- \* character c; ✓

(xxiv) The escape sequence is used for:

- \* tab
- \* backspace
- \* end of string
- \* new line ✓

(xxv) This is called the address operator:

- \* & ✓
- \* %
- \* &&
- \* |

(xxvi) The library function sqrt (1024) returns:

- \* 512
- \* 32 ✓
- \* 256
- \* 64

(xxvii) Quantities whose values cannot be changed during program execution are:

- \* data files
- \* variables
- \* constants ✓
- \* functions

(xxviii) If a = -5, the statement a += 5 returns:

- \* -15
- \* 0 ✓
- \* 10
- \* -5

(xxix) This expression is valid:

- \* a++1
- \* 6++
- \* a++ ✓
- \* a++1

(xxx) This operator operates on only one operand:

- \* =
- \* ==
- \* &&
- \* ! ✓

(xxxi) This is invalid variable declaration

- \* double c;
- \* unsigned int F;
- \* float p,q ✓
- \* char name;

(xxxii) What will be the output of the following statement:

- printf("%i, 2f", 633.6658)
- \* 63.33
- \* 63
- \* 633.66 ✓
- \* 633.6

(xxxiii) This statement can terminate any loop immediately:

- \* continue
- \* switch
- \* break ✓
- \* goto

(xxxiv) A format specifier used for int type variable is:

- \* %c
- \* %f
- \* %d ✓
- \* %s

(xxxv) Function prototype is called:

- \* Function definition
- \* Function Declaration ✓
- \* Function Calling
- \* Function Repetation

(xxxvi) This sub directory contains header files:

- \* lib
- \* bin
- \* include ✓
- \* bgi

### SECTION 'B' (Short-Answer Questions) (25)

**NOTE:** Answer any five part questions from this section. All questions carry equal marks. Answer should not exceed 3 to 4 lines.

2. i) What is IDE? Write few shortcut keys and their functions in IDE.
- ii) Write a program to print the factorial of an inputted number.

## PAST PAPERS XII (S)

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## COMPUTER (C)

iii) What is meant by Escape Sequence? Define any four with examples.

iv) Convert the following mathematical expressions into equivalent 'C' expressions. (Attempt any two)

$$a) B = (x + y)^2 - (a^2 + b^2)$$

$$b) p = \frac{(a^2 - b^2)}{(a - b^2)}$$

$$c) y = (a + b)^n$$

$$d) X = \sqrt{b + a^2}$$

v) What is main difference between Source code and Object code.

vi) Define Logical operators with examples.

vii) What are the library functions? Name any four.

viii) What are Local and Global Variables also give examples:

ix) By using nested loop write the program to print the following output.

|   |    |    |    |    |
|---|----|----|----|----|
| 2 | 4  | 6  | 8  | 10 |
| 3 | 6  | 9  | 12 | 15 |
| 4 | 8  | 12 | 16 | 20 |
| 5 | 10 | 15 | 20 | 25 |

x) Write a program to calculate the area of rectangle

(Hint  $A = w \times l$ ).

### SECTION "C" (Detailed Answer Questions) (14)

**NOTE:** Attempt any Two questions from this section. All questions carry equal marks.

3. Explain all data types in C - Language

4. How many types of decision structures are there in 'C'? Define with examples.

5. What is Function? What are the advantages of using function?

6. Define switch and break Statement with example.

نوٹ: سال 2020 میں گیارہویں اور بارہویں جماعت کے امتحانات نہیں ہوئے تھے۔

## COMPUTER 2019

Important Instructions: This paper comprises Two independent options

(I) Programming Using 'C' and (II) Programming Using Visual Basic.

The candidates have to attempt any One Option, Programming Using

'C' OR Programming Using Visual Basic, whichever they have

offered.

### OPTION - I - PROGRAMMING USING "C"

**NOTE:**

i) This section consists of 15 part questions and all are to be answered.

Each question carries 1 mark.

## PAST PAPERS XII (S) 406 COMPUTER (C)

ii) Do not copy the part questions in your answer book. Write only the answer in full against the proper number of the question and its part.

Time: 20 minutes

Max. Marks: 15

### SECTION 'A' (MULTIPLE CHOICE QUESTIONS)

1. Choose the correct answer for each from the given options:

- This object in MS-Access is used to display data in printed form:  
\*form \*Query \*Report \*Table
- Let  $a=0$ , the statement `print f("%",a++);` will print this value:  
\*0 \*1 \*2 \*3
- Primary key of another related table is called: \*Candidate key  
\*Foreign key \*Alternate key \*Composite key
- This is not a loop \*while \*for \*switch \*do-while
- This is the format specifier for "int" variable:  
\*%x \*\*%c \*\*%f \*\*%d
- This is correct statement:  
\*#include <stdio.h> \*#include <stdio>  
\*#include >stdio.h< \*#include <stdio.h>
- This primary key in a Database is:  
\*Unique \*Duplicate \*Null \*None
- This is a data type of MS-Access:  
\*int \*float \*currency \*string
- This function is used to write on to the computer screen in C:  
\*main() \*clrscr \*printf() \*getch()
- It is not an Arithmetic operator:  
\*+ \*/ \*\*% \*<=
- Passing parameters/arguments/values of a function are given within the:  
\*Square Brackets [] \*Curly Brackets {}  
\*Parenthesis () \*Pair of Slashes /\*\*/
- Compiler cannot detect this error:  
\*Runtime \*Logical \*Absolute \*Syntax
- The scope of this type of variable is not limited:  
\*local variable \*Integer variable \*external/global variable  
\*float variable
- This object in MS-Access is considered as primary object:  
\*Query \*Form \*Report \*Table
- This keyword is used to exit from the switch statement:  
\*case \*break \*else \*default

## COMPUTER 2019

Option-1 Programming using 'C'

TIME: 2 Hours 40 Minutes

SECTION 'B' (SHORT ANSWER - QUESTION) (60 Marks)

SECTION 'C' (DETAILED-ANSWER QUESTIONS) (36 Marks)

## PAST PAPERS XII (S) 407 COMPUTER (C)

**NOTE:** Answer any 12 part questions from this section. All questions carry equal marks.

- (i) What is programming error? Define Syntax error.
- (ii) Write a C program to show the table of an inputted number.
- (iii) Write C-Program equivalent statements of any three of the following:  
(a)  $A = (x^2 + y^3) - (x - y)^3$  (b)  $B = y^2 - 2xy + x^2$   
(c)  $C = \frac{-y \pm \sqrt{y^2 - 4xz}}{2x}$  (d)  $D = x^2 + y^3 + z^3 - 3xyz$
- (iv) Write the basic structure of C programming language.
- (v) Differentiate between while loop and do-while loop.
- (vi) Define Relational Operators used in C-language.
- (vii) What is escape sequence? Define any two
- (viii) Write the full form of the following:  
\*DML \*RDBMS \*DBA
- (ix) What is difference between Primary key and Foreign key?
- (x) Can reserved words in C-language be chosen as variable names? Justify your answer.
- (xi) Write any three advantages of DBMS.
- (xii) Write a function that receives two integers and returns the maximum of the two.
- (xiii) What is the role of DBA in DBMS?
- (xiv) What will be the output of the give programing statements?  
For ( int x=1; x<=5; x++)  
{for (int y=1; y<=x; y++)  
Printf("%d",y);  
Print f("\n");  
}
- (xv) Identify valid variables or invalid variables of the following:  
(a) number (b) You & me (c) Case
- (xvi) Rewrite the given statements correctly:  
a) If (a=b) then  
printf("Karachi");  
b) for (int i=1, i<10, i++)  
printf("%d\n",i)  
c) integer i=5;  
i++;
- (xvii) Write the steps to create a table in MS-Access
- (xviii) Define any three reserved words in C.

### SECTION 'C' (DETAILED-ANSWER QUESTIONS) (24)

**NOTE:** Attempt Two questions from this section.

- (a) Explain all data types in C-Language.
- (b) Define "Nested" for loop with example.
- (c) Define any four objects of MS-Access.

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4. (a) What are the input/output statements in C programming language? Define any three with examples.  
 4. (b) Define else if statement. How it is different from switch statement?  
 4. (c) Write the names of various types of database models. Explain two of these.  
 5. (a) What is function in C-language? In how many ways can be used?  
 5. (b) What are control structure in C-language? Define Iteration control structure. Give example.  
 5. (c) What is relationship Database? Define all three types of relationship Database. Give example.

**COMPUTER 2018**

**OPTION - I - PROGRAMMING USING "C"**

Time: 20 minutes

Max. Marks: 15

**SECTION 'A' (MULTIPLE CHOICE QUESTIONS)**

1. Choose the correct answer for each from the given options:

- (i) A character constant is a single character, enclosed:

- \*in double quotation marks
- \*in single quotation marks
- \*in left and right braces
- \*without quotation marks

- (ii) Comments or remarks can be placed anywhere within a program as long as they are placed within delimiters:

- \*by braces and asterisks/\*comments\*/
- \*by asterisks \*comments\*
- \*by parenthesis and asterisks/\*comments\*/
- \*by slashes and asterisks /\* comments \*/

- (iii) The function void play (void); is the example of:

- \*Calling function
- \*A Prototype declared as a function
- \*Return type function
- \*Itself function

- (iv) This operator has the lowest precedence:

- \*++
- \*%
- \*<
- \*&&

- (v) This converts C program into machine language:

- \*An editor
- \*A compiler
- \*An operating system
- \*A linker

- (vi) This function is used to read a character from a data file:

- \*putchar()
- \*getc()
- \*getch()
- \*getche()

- (vii) Loop within the loop is called:

- \*Nested loop
- \*Continuous loop
- \*Inner loop
- \*Outer loop

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- (viii) This database tool is used to display and print your data in an organized manner: \*Record \*Field \*Report \*Table

- (ix) This database model organizes data into a tree structure:

- \*Hierarchical
- \*Relational
- \*Network model
- \*Object oriented model

- (x) Each row in a table is also called a:

- \*Field
- \*Tuple
- \*Entity
- \*Attribute

- (xi) A person who is responsible for managing the overall database management system is called :

- \*Database administrator
- \*Network supervisor
- \*System analyst
- \*Programmer

- (xii) This is escape sequence: \n \*1 \*%f \*j--

- (xiii) This is the output of 4% 5: \*0 \*1 \*4 \*5

- (xiv) The specialized software necessary to create and process the database and modified as needed is called:

- \*Compiler
- \*Windows xp
- \*DBMS
- \*Interpreter

- (xv) The include <stdio.h> does not begin without a sign of:

- \*\$
- \*!=
- \*#
- \*@

**COMPUTER 2018**

Option-I Programming using 'C'

TIME: 2 Hours 40 Minutes

(60 Marks)

**SECTION 'B' (SHORT ANSWER - QUESTION) (36 Marks)**

**NOTE:** Answer any 12 questions from this section.

2. (i) If x=1, what is the output of following program segments?

```
Printf("x=%d\n",x);
Printf("x=%d\n",++x);
Printf("x=%d\n",x);
```

2. (ii) What is the difference between source program and an object program?

2. (iii) Determine which of the following are valide or invalid identifiers.

- (a) Array1 (b) Stax (c) name and address

2. (iv) What is the difference between arithmetic and arithmetic assignment operators?

2. (v) What is the difference between DBMS and RDBMS?

2. (vi) Write the full form of DCL, IDE and DBMS.

2. (vii) Write the name of objects in MS-Access.

2. (viii) Define the following functions:

- (a) Pow() (b) sqrt() (c) abs()

2. (ix) Write the print () statements of the following:

- (a) Int a=4562; (b) float c=15.27; (c) char t[]

- "intermediate class";

2. (x) Write the following in while () equivalent:

Void main (void)

```

int a=3;
do {printf("%d\n",a);
 A=a+3;
}

```

While (a=10);

2. (xi) Define Relational database model.

2. (xii) Convert the following algebraic expression to C-equivalent format:

$$(a)x = \frac{1}{2}ab^2 \quad (b)x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \quad (c)y = |b^2 - 4ac|$$

2. (xiii) Write a C-program to calculate the power of numbers from 1 to 5.

2. (xiv) What is the meaning of two void used in any function declaration?

i.e. void main(void)

2. (xv) What is wrong with the following C-program?

```

void main(void)
{

```

```

 Print Hoo what a wonderful short
}

```

2. (xvi) Write a program that prints factorial of any inputted member.

2. (xvii) What is composite key? Where do we use it?

2. (xviii) How can we define one-to-one relationship in database?

### SECTION 'C' (DETAILED-ANSWER QUESTIONS) (24)

**NOTE:** Answer Two questions from this section. All questions carry equal marks.

3 (a). Define switch and break statement with an example.

3 (b). Write the general form(syntax) of for () & while () loops.

3 (c). Explain why we use Table, Form & Report in MS- Access.

4 (a). What are the different types of operators used in any C Language program? Define any two types.

4 (b). Define different basic opening modes used to open a data file in C language.

4 (c). Why is DBMS necessary for any big organization?

5 (a). Define Function. What are the main characteristics of a function.

5 (b). Define different format specifiers used in any formatted function.

5 (c). Define the data types in MS access.

## COMPUTER 2017

### OPTION - I - PROGRAMMING USING "C"

Time: 20 minutes

Max. Marks: 15

### SECTION 'A' (MULTIPLE CHOICE QUESTIONS)

1. Choose the correct answer for each from the given options:

i) This format specifier is used for double precision value:

\* %ld      \* %lf      \* %d      \* %u

ii) The quantities which may change during program execution are called:

\* Constants      \* Variables      \* Strings      \* Integers

iii) The 'break' statement is used to exit from this part of a loop:

\* Beginning      \* Middle      \* End      \* Never exit

iv) This is the default data type in MS Access:

\* Text      \* Auto number      \* Number      \* Data

v) The "\n" character does this one of the following operations:

\* double line spacing      \* character deletion

\* character backspace      \* place cursor on the next line

vi) # define PI 3.14159. This code represents this one of the following:

\* Preprocessor directive      \* Variable

\* Constant      \* Integer

vii) This arithmetic operator computes the integer remainder:

\* %      \* +      \* -      \* /

viii) A top to bottom relationship among the items in a data base is established by a:

\* Hierarchical Model      \* Network Model

\* Relation Model      \* Flat Model

ix) This one of the following represents comment in C:

\* <      \* ||      \* /\* \*/      \* /\* \*/

x) A relational database developer refers to record as:

\* a criteria      \* a relation      \* a tuple      \* an attribute

xi) The format specifier starts with:

\* %      \* /      \* \      \* #

xii) This MS Access object is considered primary:

\* Table      \* Query      \* Form      \* Report

xiii) "=" is an operator of this type:

\* Logical      \* Arithmetic      \* Assignment      \* Relation

xiv) Reserved word "case" is used with this statement:

\* printf()      \* scanf()      \* switch()      \* getch()

xv) This standard I/O function is used to close a data file:

\* gclose()      \* fclose()      \* fscanf()      \* fcall()

## COMPUTER 2017

Option-I Programming using 'C'

TIME: 2 Hours 40 Minutes

(60 Marks)

### SECTION 'B' (SHORT ANSWER - QUESTION) (36 Marks)

**NOTE:** Answer any 12 questions from this section.

2. (i) What is foreign key? Where do we use it?

2. (ii) What is the main difference between while and do-while loop?

2. (iii) In which condition is one to many relationship used?

2. (iv) What is the output of the following program statements:

```

For (int a=1; a<5; a++)
{

```

```

For (int b=a; b<5; b++)
 Printf ("%d",B);
 Printf ("\n");
}

```

2. (v) What do you mean by function prototype?  
 2. (vi) What is operator precedence? Write the precedence arithmetic operators.  
 2. (vii) Write a program that swap two inputted numbers.  
 2. (viii) Write the full form of the following:  
 \*SQL \*DML \*DDL  
 2. (ix) How does else-if ladder differ from switch statement?  
 2. (x) What is data dictionary in DBMS.  
 2. (xi) What will be the output of the following program:  
 Void main()

```

{
 int n=5;
 if (n<1)
 Printf ("hello");
 if (n==5)
 Printf ("hi")
 else
 printf("no");
}

```

2. (xii) Write the equivalent code of the following statement in while loop format:

```

for(a=1; a<=100; a++)
 Printf ("%d\n", a*a);

```

2. (xiii) Define an increment or decrement statement in C.  
 2. (xiv) Define the following file opening mode in C:  
 (a) "r" (b) "w" (c) "a"  
 2. (xv) Debug the following program:

```

{
 Printf ("I am a good programmer");
}

```

2. (xvi) Write a program to calculate the area of a rectangle (Hint  $A = w \times h$ )  
 2. (xvii) What do you mean by reserved words in C language?  
 2. (xviii) Write equivalent 'C' statement:

$$(a) x = \frac{b^2 + c^2}{2bc} \quad (b) b = \sqrt{x + y + z^2} \quad (c) z = \frac{4}{3}\pi r^3 \text{ where } \pi = 3.14$$

### SECTION 'C' (DETAILED-ANSWER QUESTIONS) (24)

**NOTE:** Answer Two questions from this section. All questions carry equal marks.

- 3 (a). What is meant by Control Structure?. Discuss selection control structure  
 3 (b). Define Database. Define the major data types used in MS access

- 3 (c). What is the purpose of break and continue statement?  
 4 (a). Define the data types of C.  
 4 (b). Define Network Model. Where do we use it?  
 4 (c). Define formatted I/O and unformatted I/O functions.  
 5 (a). Define any four objects of DBMS.  
 5 (b). Define for loop with an example.  
 5 (c). What is Function? Define its importance in C language.

## COMPUTER 2016

### OPTION - I - PROGRAMMING USING "C"

NOTE:

- i) This section consists of 15 part questions and all are to be answered. Each question carries 1 mark.  
 ii) Do not copy the part questions in your answer book. Write only the answer in full against the proper number of the question and its part.

Time: 20 minutes

Max. Marks: 15

### SECTION 'A' (MULTIPLE CHOICE QUESTIONS)

1. Choose the correct answer for each from the given options:

- i) Format specifiers must start with: \* % \* & \* \* \*  
 ii) This is a relational expression:  
 \*  $x = y$ ; \*  $x \neq y$ ; \*  $x + y$ ; \*  $x \rightarrow$   
 iii) The gets () statement is used to input:  
 \* Single character data \* Integer data  
 \* String data \* Float data  
 iv) To print the value of %f a and b given below, the following printf() statement will be used: float a = 3.14; double b = 2.532;  
 \* printf ("%f%f", a, b); \* printf ("%Lf%f", a, b);  
 \* printf ("%Lf%Lf", a, b); \* printf ("%f%Lf", a, b);  
 v) The columns in a Microsoft Access Table are called:  
 \* Rows \* Records \* Fields \* Entities  
 vi) A foreign key:  
 \* has nothing to do with the primary key  
 \* has different values than the primary key  
 \* is found in tables that do not have a primary key  
 \* is related to the primary key of a different table  
 vii) The various components in MS Access are called:  
 \* Tools \* Entities \* Properties \* Objects  
 viii) A field in DBMS is sometimes called a/an:  
 \* File \* Attribute \* Record \* Database  
 ix) A function which invokes itself is called:  
 \* Recursive function \* System function  
 \* Library function \* Iteration function  
 x) Names given to variable, constants and functions are called:

- \* header files
- \* identifiers
- \* loops
- \* structures
- xi) For () loop is used when we need to perform:
  - \* Sequential logic
  - \* Selection logic
  - \* Iteration logic
  - \* Parallel logic
- xii) This is a statement terminator in C.:
  - \* {
  - \* }
- xiii) This type of error is detected by the compiler:
  - \* Run time
  - \* Logical
  - \* Syntax
  - \* Absolute
- xiv) A pictorial / graphical representation of program logic is known as:
  - \* Flowchart
  - \* Procedure
  - \* Source code
  - \* Algorithm
- xv) A program, written in high level language, is known as:
  - \* Source code
  - \* Object code
  - \* Uni code
  - \* Executable code

## COMPUTER 2016

Option-I Programming using 'C'

TIME: 2 Hours 40 Minutes

(60 Marks)

### SECTION 'B' (SHORT ANSWER - QUESTION) (36 Marks)

**NOTE:** Answer any 6 questions from this section. All questions carry equal marks.

2. (i) Define each format specifier with an example  
(a) %c (b) %d (c) %f (d) %s (e) %lf (f) %ld
2. (ii) Determine any six valid or invalid identifiers, in case of invalid, given reason.  
(a) Include (b) Stwenty (c) S-numbers (d) y/s (e) integer (f) \_name (g) Getch (h) smallest
2. (iii) Draw a flowchart that prints the table of any inputted number.
2. (iv) What is meant by case-sensitive language? Write the structure of "C" program.
2. (v) Write a program in C language to print the number from 4 to 9 and their squares. (using do while loop).
2. (vi) Convert the following mathematical expressions into "C" equivalent statements.  
(a)  $A = \pi r^2$  (b)  $x = 3ab^3 + 3a^2b$  (c)  $d = |b^2 - 4ac|$  (d)  $y = a^2 + b^3 + c^4$  (e)  $X = (a^2 - b^3)^n$  (f)  $J = \frac{1}{2}pb$
2. (vii) Write the output of the following statement of C-program.  

```

For(int i=2; i<10; i++)
{
 Printf("%d", i);
 If(i%2==0)
 Sontinue;
 Printf("\n");
}

```

2. (viii) What is algorithm? Define different Algorithm Notations:
2. (ix) What are the responsibilities of DBA?

### SECTION 'C' (DETAILED-ANSWER QUESTIONS) (24)

**NOTE:** Answer Two questions from this section. All questions carry equal marks.

- 3 (a). What is loop? Explain nested for loop with an example.
- 3 (b). What is preprocessor directive? Explain any two preprocessor directive in C with examples.
- 4 (a). Define fails. Describe any two types of data files using in C-program.
- 4 (b). Define operators. Explain different types of operators in C-language.
- 5 (a). Explain the objects of MS Access.
- 5 (b). What is database? What are the advantages of relational database management system (RDMBS)?

## COMPUTER 2015

### OPTION - I - PROGRAMMMING USING "C"

Time: 20 minutes

Max. Marks: 15

### SECTION 'A' (MULTIPLE CHOICE QUESTIONS)

1. Choose the correct answer for each from the given options:

- i) The for loop counter variables should be of this type:
  - \* Float
  - \* int
  - \* Char
  - \* Double
- ii) A field is sometimes called an/a:
  - \* File
  - \* Attribute
  - \* Record
  - \* Database
- iii) An object whose value can be changed during the program execution is called:
  - \* Symbol
  - \* Variable
  - \* Constant
  - \* Operator
- iv) To control the spacing and number of characters is displayed by:
  - \* Precision specifier
  - \* Field-width specifier
  - \* Format specifier
  - \* Escape sequence
- v) This database object is used to present data in a printed format:
  - \* Table
  - \* Query
  - \* Form
  - \* Report
- vi) The memory space required for a float is:
  - \* 2 bytes
  - \* 8 bytes
  - \* 1 byte
  - \* 4 bytes
- vii) A function which invokes itself is called:
  - \* Recursive function
  - \* System function
  - \* Library function
  - \* Iterative function
- viii) In database, records are also called:
  - \* Attributes
  - \* Entities
  - \* Tuples P
  - \* Relations
- ix) This is the symbol of AND operator:
  - \* %&
  - \* &&
  - \* &
  - \* !
- x) The control structure use do-while loop is:
  - \* Sequence
  - \* Selection
  - \* Iteration
  - \* Random
- xi) This symbol in a flow chart represents a process: